# Lecture Scribe Request

> "Generate a lecture scribe intended to serve as reference material for exam preparation. Use *ONLY* the provided context (Lecture 5 slides).
> *Instructions:*
> 1. *Purpose:* The scribe must be exam-ready, strictly following the content of the slides.
> 2. *Content:* Include definitions, assumptions, theorem statements, proofs or derivations, and worked examples *exactly* as presented in the lecture materials.
> 3. *Constraints:* Do not introduce new examples, outside information, or hallucinations. Do not use phrases like 'for example' to add your own content.
> 4. *Structure:* Organize the scribe clearly with appropriate headings corresponding to the lecture flow.
>
>
>
> *Goal:* Produce a clean, accurate revision document based solely on the attached file."

Got it 👍 I'm ready to generate the **exam-ready lecture scribe** exactly as you specified.

One thing first: I don't see the **Lecture 5 slides** attached yet.

Please **upload the Lecture 5 slides (PDF/images)**, and once they're attached I will:

- Stick **strictly** to the slide content
- Include **only** the definitions, assumptions, theorems, derivations/proofs, and examples **as presented**
- Follow the **same flow and structure** as the lecture
- Avoid adding *any* external explanations, examples, or assumptions
- Make it **clean, precise, and revision-focused** for exams

Upload the file and I'll start immediately 📘✨

wait i will attach it

All good 👍
Take your time — once you attach the **Lecture 5 slides**, I'll start preparing the **exam-ready lecture scribe** strictly based on them, exactly as per your instructions.

**L5_A_S2.pdf**
PDF

here

Below is the **exam-ready Lecture Scribe for Lecture 5**, prepared **strictly and only** from the provided slides, following the **same flow, definitions, derivations, and examples** as presented. No external material or additional examples have been introduced.
Source: **Lecture 5 slides (CSE400)**

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# CSE400 – Fundamentals of Probability in Computing

## Lecture 5: Bayes' Theorem, Random Variables, and Probability Mass Function

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## 1. Bayes' Theorem

### 1.1 Weighted Average of Conditional Probabilities

Let $A$ and $B$ be events.

We may express event $A$ as:

$$A = AB \cup AB^c$$

This is because, for an outcome to be in $A$, it must either:

- be in both $A$ and $B$, or
- be in $A$ but not in $B$.

Since $AB$ and $AB^c$ are mutually exclusive, by **Axiom 3**:

$$\Pr(A) = \Pr(AB) + \Pr(AB^c)$$

Using conditional probability:

$$\Pr(A) = \Pr(A \mid B)\Pr(B) + \Pr(A \mid B^c)[1 - \Pr(B)]$$

**Interpretation:**
The probability of event $A$ is a **weighted average of conditional probabilities**, where the weights are the probabilities of the events on which we condition.

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## 1.2 Learning by Example – Example 3.1 (Part 1/2)

**Problem Statement:**
An insurance company divides people into two classes:

- Accident-prone
- Not accident-prone

Statistics show:

- An accident-prone person has an accident within one year with probability $0.4$.
- A non-accident-prone person has an accident within one year with probability $0.2$.
- $30\%$ of the population is accident-prone.

Find the probability that a new policyholder will have an accident within one year.

**Solution:**
Let:

- $A_1$: policyholder has an accident within one year
- $A$: policyholder is accident-prone

Then,

$$\Pr(A_1) = \Pr(A_1 \mid A)\Pr(A) + \Pr(A_1 \mid A^c)\Pr(A^c)$$

$$= (0.4)(0.3) + (0.2)(0.7) = 0.26$$

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## 1.3 Learning by Example – Example 3.1 (Part 2/2)

**Problem Statement:**
Suppose a new policyholder **has** an accident within a year.

Find the probability that the policyholder is accident-prone.

**Solution:**

$$\Pr(A \mid A_1) = \frac{\Pr(AA_1)}{\Pr(A_1)} = \frac{\Pr(A)\Pr(A_1 \mid A)}{\Pr(A_1)}$$

$$= \frac{(0.3)(0.4)}{0.26} = \frac{6}{13}$$

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### 1.4 Formal Introduction: Law of Total Probability

Suppose:

- $B_1, B_2, \ldots, B_n$ are **mutually exclusive events**
- $\bigcup_{i=1}^{n} B_i = B$

Exactly one of the events $B_1, B_2, \ldots, B_n$ must occur.

Writing:

$$A = \bigcup_{i=1}^{n} AB_i$$

Since $AB_i$ are mutually exclusive:

$$\Pr(A) = \sum_{i=1}^{n} \Pr(AB_i) = \sum_{i=1}^{n} \Pr(A \mid B_i)\Pr(B_i)$$

This is called the **Law of Total Probability (Formula 3.4)**.

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### 1.5 Bayes Formula (Proposition 3.1)

Using:

$$\Pr(AB_i) = \Pr(B_i \mid A)\Pr(A)$$

We obtain:

$$\Pr(B_i \mid A) = \frac{\Pr(A \mid B_i)\Pr(B_i)}{\sum_{j=1}^{n} \Pr(A \mid B_j)\Pr(B_j)}$$

Where:

- $\Pr(B_i)$: **a priori probability**
- $\Pr(B_i \mid A)$: **posteriori probability**

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### 1.6 Learning by Example – Example 3.2 (Card Problem)

Three cards:

- Card 1: red–red
- Card 2: black–black
- Card 3: red–black

One card is randomly selected and placed on the ground.
The upper side is red.

Find the probability that the other side is black.

**Solution:**
Let:

- $RR$: all-red card
- $BB$: all-black card
- $RB$: red–black card
- $R$: upturned side is red

$$\Pr(RB \mid R) = \frac{\Pr(R \mid RB)\Pr(RB)}{\Pr(R \mid RR)\Pr(RR) + \Pr(R \mid RB)\Pr(RB) + \Pr(R \mid BB)\Pr(BB)}$$

$$= \frac{(1/2)(1/3)}{(1)(1/3) + (1/2)(1/3) + (0)(1/3)} = \frac{1}{3}$$

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## 2. Random Variables

### 2.1 Motivation and Concept

When an experiment is performed, interest often lies in a **function of the outcome**, not the outcome itself.

Examples from slides:

- In dice tossing, focus is on the **sum** of dice.
- In coin tossing, focus is on the **number of heads**.

**Definition:**
A **random variable** is a real-valued function defined on the sample space.
Values are determined by outcomes, and probabilities are assigned to these values.

Formally:

$$X : \Omega \to \mathrm{R}$$

The lecture restricts attention to **discrete random variables**, whose values form a finite or countably infinite subset of $\mathrm{R}$.

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## 2.2 Distribution of a Random Variable

Two key components:

1. The set of values the random variable can take
2. The probabilities with which it takes those values

For any value $a$:

$$\{\omega \in \Omega : X(\omega) = a\}$$

is an event.

The probability:

$$\Pr(X = a)$$

defines the **distribution** of the random variable.

The distribution can be visualized as a **bar diagram**, where:

- x-axis: values of the random variable
- bar height: $\Pr(X = a)$

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## 2.3 Example – Tossing 3 Fair Coins

Let $Y$ be the number of heads.

Possible values: $0, 1, 2, 3$

$$\Pr(Y = 0) = \frac{1}{8}, \quad \Pr(Y = 1) = \frac{3}{8}, \quad \Pr(Y = 2) = \frac{3}{8}, \quad \Pr(Y = 3) = \frac{1}{8}$$

Since $Y$ must take one of these values:

$$1 = \sum_{i=0}^{3} \Pr(Y = i)$$

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## 3. Probability Mass Function (PMF)

### 3.1 Concept

A random variable that can take on **at most a countable number of values** is called **discrete**.

Let $X$ be a discrete random variable with range:

$$R_X = \{x_1, x_2, x_3, \ldots\}$$

The function:

$$p(x_k) = \Pr(X = x_k)$$

is called the **Probability Mass Function (PMF)** of $X$.

Since $X$ must take one of these values:

$$\sum_k p(x_k) = 1$$

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### 3.2 PMF Example

Given:

$$p(i) = c\frac{\lambda^i}{i!}, \quad i = 0, 1, 2, \ldots$$

Find $\Pr(X = 0)$ and $\Pr(X > 2)$.

Since:

$$\sum_{i=0}^{\infty} p(i) = 1 \Rightarrow c \sum_{i=0}^{\infty} \frac{\lambda^i}{i!} = 1$$

Using:

$$e^{\lambda} = \sum_{i=0}^{\infty} \frac{\lambda^i}{i!}$$

We get:

$$ce^{\lambda} = 1 \Rightarrow c = e^{-\lambda}$$

Thus:

$$\Pr(X = 0) = e^{-\lambda}$$

$$\Pr(X > 2) = 1 - \Pr(X \leqslant 2) = 1 - [\Pr(X = 0) + \Pr(X = 1) + \Pr(X = 2)]$$

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## End of Lecture 5

**Class participation via Campuswire**

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If you want, next I can:

- Convert this into **LaTeX exam notes**
- Compress it into a **last-day revision sheet**
- Create **exam-oriented Q&A strictly from these slides**